



Kairos

Knowledge that arrives before you ask for it.

PROBLEM STATEMENT

Industrial knowledge that never reaches the point of decision.

Artificial Intelligence & Machine Learning

Computer Vision

Generative AI

TEAM

Deterium

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The plant already knows. Nobody can find it in time.

At 2:40 a.m. a technician signs a permit to open a feed pump. The approved procedure in front of him is current, correct, and wrong for this pump. The bulletin that lowered the limit is in the building, in another system, under another name for the same equipment.

He did not fail to find it. He failed to know it existed.

Over 90%

of the serious incidents the US Chemical Safety Board analysed involved hazards already documented in publicly available literature.

Incident Data: Reactive Hazard Investigation, No. 2003-15-D. 167 US incidents, 1980 to 2001.

HIT BY AN UNPLANNED OUTAGE AT LEAST MONTHLY

India 88%

Global 69%

Unplanned downtime also costs ₹70 lakh an hour in India, against ₹1.03 crore globally. ABB Value of Reliability, Sapio Research, July 2023, n=3,215.

Events trigger the knowledge, not questions

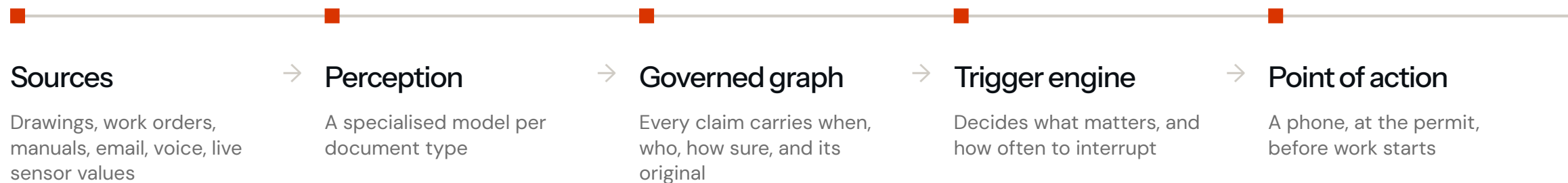
EVERY SYSTEM BEFORE THIS

Knowledge exists → someone forms a query → results come back

KAIROS

An event happens → the system reads it against everything the plant knows → the brief arrives, unasked

Nobody types anything. A work order opening, a permit raised, a shift handing over, equipment isolated: each is a trigger.



A system that pushes must not become noise, so interruptions are capped at the process industry's own alarm benchmark: fewer than six per operator per hour (EEMUA 191, ANSI/ISA-18.2, IEC 62682).

Six things it does that a search box cannot

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- | | |
|--|--|
| ■ Briefs that arrive unasked | A work order opening, a permit raised, a shift handing over: each is a trigger. Nobody searches. |
| ■ Every answer carries its evidence | Source document, authority level, confidence score, and a link to the original file, on every claim. |
| ■ A refusal instead of a guess | Below the evidence threshold on a safety question it states what is missing, never a hedged answer. |
| ■ Time travel over the record | Ask what the plant knew on the day of the incident, not what it knows after the investigation. |
| ■ Conflict detection and blast radius | When a source is superseded, everything downstream of it is traced and flagged for re-review. |
| ■ Compliance gaps by clause | Each regulatory clause checked against the documents actually linked to that asset, and the gaps listed. |
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The first three are the product. Delivery without governance is a confident guess; governance without delivery is a better document store.

Two claims, and the four mechanisms underneath

The inversion. Every prior approach optimises retrieval, and every one assumes somebody knows what to ask. The most dangerous gaps are the ones nobody thinks to query.

01 Every fact knows when it stopped being true

02 A regulation outranks a field note

03 Every answer points at the original file

04 It is built to refuse

The screenshot displays a user interface with a dark background. At the top, a question is posed: "What changed on P-101 since the last turnaround?". Below this, a white box labeled "CHECKED ANSWER" contains the text: "The pressure limit dropped to 16.2 bar on 14 May." Underneath the answer, there are three horizontal bars of varying lengths. To the left of these bars is a red box containing the text "3 sources". To the right is a grey box containing the text "Authority: regulation". Below these elements, the word "Confidence" is followed by a red progress bar that is nearly full, and the number "0.91". At the bottom of the interface, a dashed white box contains a small square icon followed by the text: "Thin evidence on a safety question returns a **refusal card**, with its sources, never a hedge."

Without proactive delivery this is a better document store. Without governed accuracy it is a confident hallucination machine pointed at a refinery.

Five roles, each reached at one moment



Field technician

At the permit, on a phone,
one tap to acknowledge.
Never opens a search box.



Reliability engineer

At the failure, with the
equipment's history
already assembled.



Shift lead

At handover, inheriting a
structured brief instead of
what someone
remembered.



Compliance officer

Before an audit, with
evidence organised by the
clause it answers to.



Plant head

Across sites, where a
failure at one plant warns
the other five.

Bought by plant operations and reliability leadership, who carry both the downtime budget and the safety accountability. Deployed first in Indian heavy industry, where multilingual and mixed-script field records are the norm rather than an edge case.

What we will measure, not what we will claim

We will not project a percentage we cannot yet evidence.

The binding constraint was never search speed. It is that the right knowledge does not reach the right person at the moment of the decision: the permit, the work order, the handover.

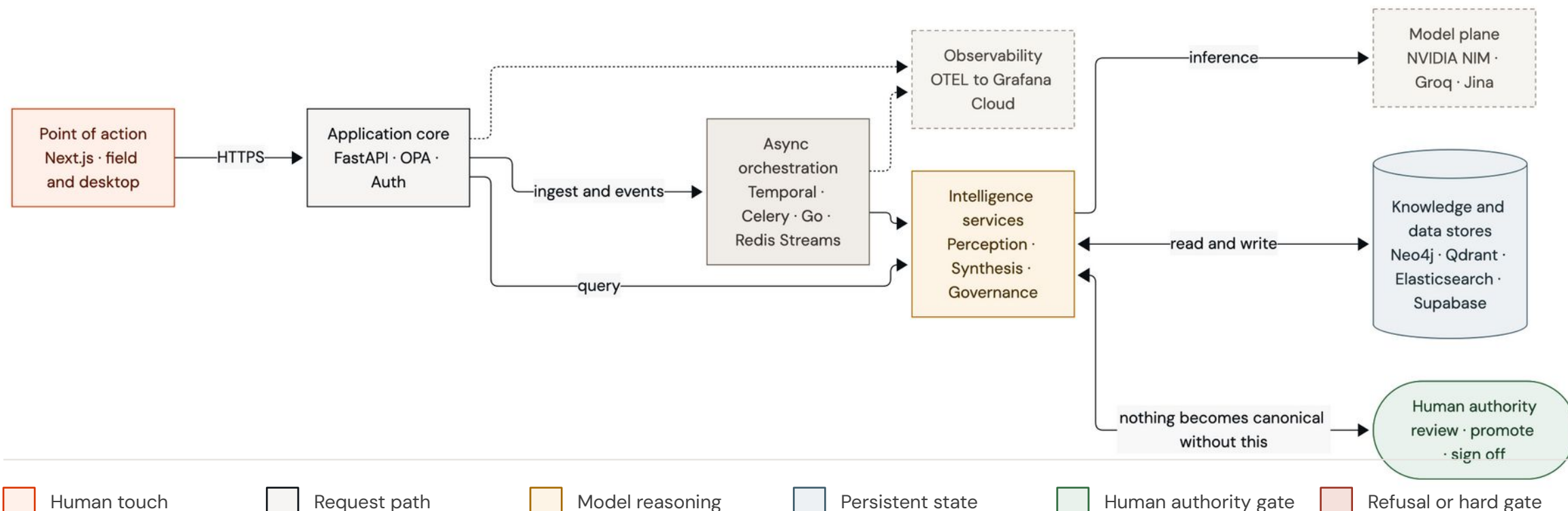
FOUR INSTRUMENTS, BUILT ALONGSIDE THE SYSTEM

- 01 Answer correctness against a fixed expert question set with a written answer key
- 02 Time to a trusted answer, against plain keyword search on the same corpus
- 03 Compliance gap detection, scored on recall first: a missed gap is a liability, a false alarm a nuisance
- 04 Proactive brief relevance, against what the receiving role actually needed

Graded deterministically against a written key. Never one model scoring another model's output.

A store per job, a model per task

Every box is a decision, not a vendor list: a graph store because the queries are multi-hop traversals filtered by time and authority, exact search alongside semantic because equipment tags must match exactly, and a specialist model per document type because one generalist degrades on all of them.



Green and brick appear only on the human gate and the refusal path: the two places the system declines to act on its own. All inference runs on hosted APIs, so there is no GPU procurement, no training run and no labelled dataset on the critical path.

Built to survive contact with a real plant

The question is not whether a demo can be assembled. It is whether this can be implemented beyond the hackathon, inside a brownfield plant that will not stop running while you install it.

WHY IT IS IMPLEMENTABLE

- All inference runs on hosted APIs: no GPU procurement, no training run, no labelled dataset
- Brownfield entry. It reads the systems a plant already runs; nothing is ripped out or replaced
- Value lands before the graph does. Month one is search across everything ingested
- The human gate lets it enter a safety-critical environment without waiting for perfect extraction
- Every store is a managed or containerised service with a well-understood operating model

CONSTRAINTS WE ARE NOT HIDING

Graph query performance as the graph grows. Index design is a workstream from month two, not late hardening

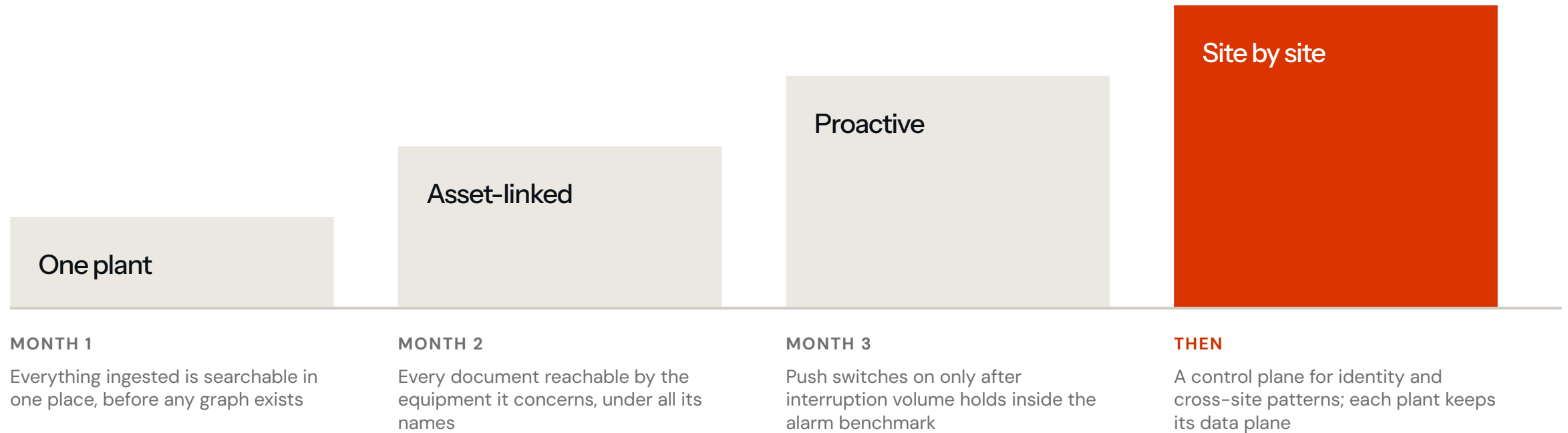
Drawing extraction is candidate, not canonical, until an engineer verifies it element by element

Benchmark figures will characterise a seeded corpus, not a production archive

Compliance output is audit preparation with human sign-off, never automated compliance

For the prototype round: one vertical slice end to end, ingest through to a cited answer and one proactive brief, measured against a written answer key. A prototype that does one thing all the way through beats five that stop at the seam.

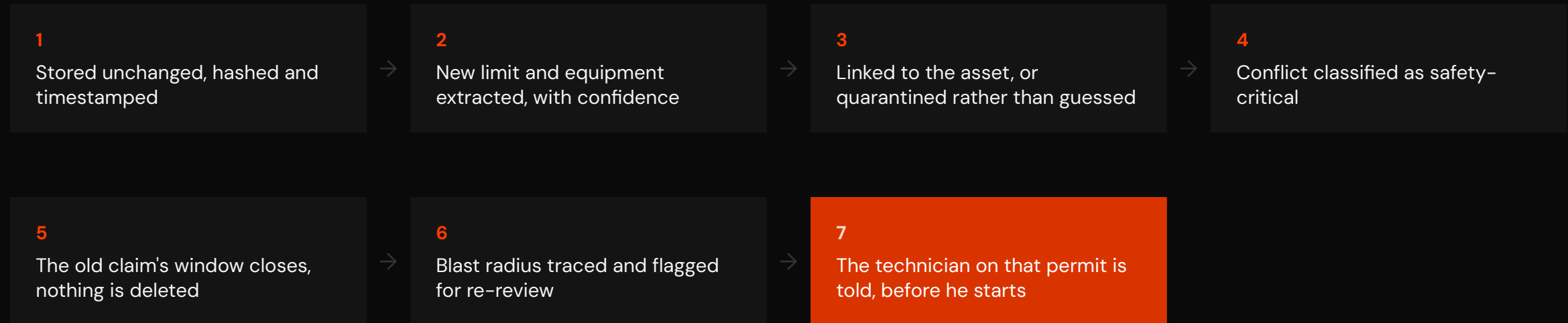
One plant, then every plant, then every sector



Data stays inside the plant. Only an anonymised technical pattern crosses a site boundary, with personnel names stripped. Across sectors the architecture is identical: regulations, drawing conventions and vocabulary are configuration.

One event, traced end to end

A manufacturer's bulletin arrives lowering a pressure limit.



Step 7 is the product. Steps 1 to 6 are why step 7 can be trusted.

Phase 1 corpus, storage, ingestion · Phase 2 graph and retrieval · Phase 3 synthesis, refusal gate, trigger · Phase 4 field view, evaluation, documentation

Evaluation is built in phase 1, not phase 4. A measured result is the deliverable, not a demonstration.

